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Edge Detection

Image processing methods
Kamil Musiał, PhD, Eng.
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What an edge detector tells us

An edge detector finds places where image brightness changes strongly over a short distance. It converts a rich image into a simpler description of local structure.

Useful applications include finding outlines, checking alignment, reading document structure and preparing shape measurements. A detector reports image evidence; it does not identify the meaning of an object.

A good result keeps relevant boundaries, places them accurately and avoids unnecessary responses. The desired result depends on the task.

Brightness edges and object boundaries

Camera photograph | 512 x 512 px



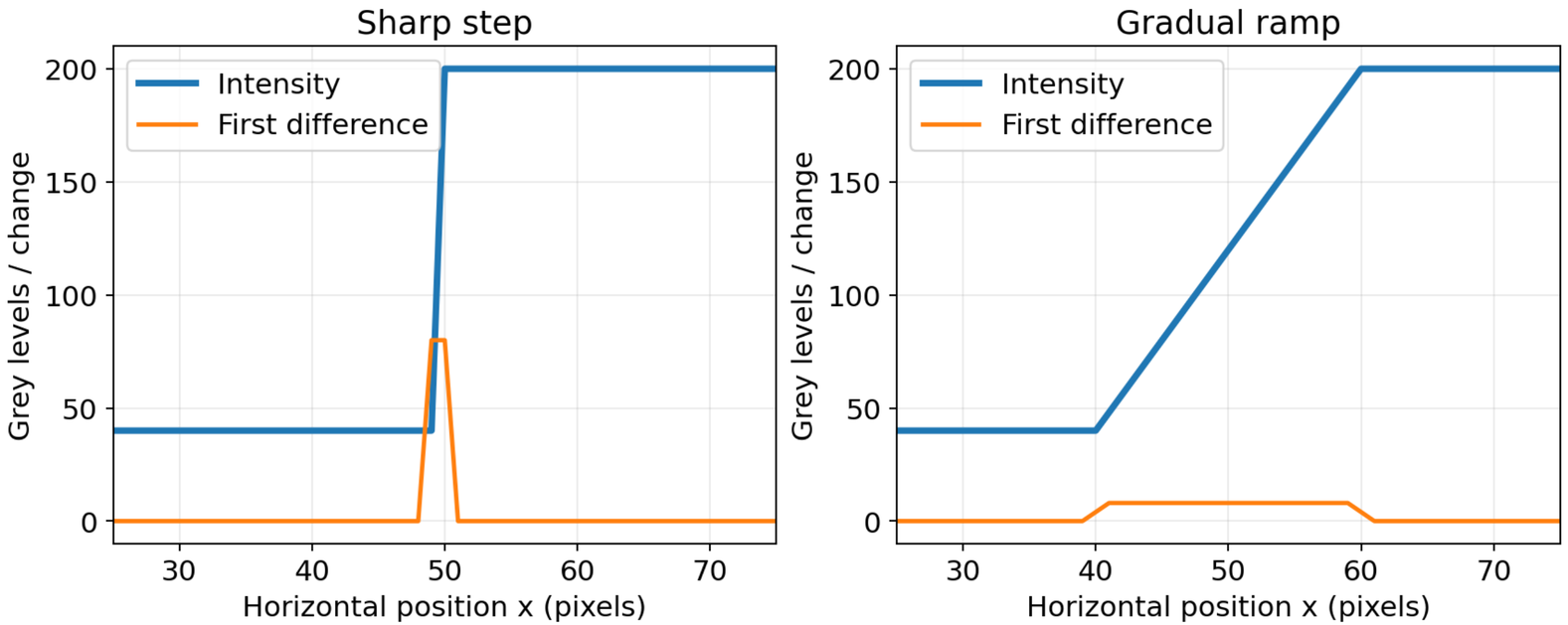
Canny edges in orange



Orange pixels mark selected brightness changes. Some follow the person or camera; others follow texture, shadows and internal details.

A physical boundary can be weak or missing. An edge map gives candidate structures, not a complete object mask.

A step and a ramp



A sharp step changes from dark to bright over a small distance. Its first difference has a narrow peak.

A ramp spreads the change across more pixels: the response is smaller and wider. Blur changes the profile without moving the physical boundary.

A derivative means local change

A simple horizontal difference subtracts two neighbouring brightness values. For the row [40, 40, 200, 200], the forward differences are [0, 160, 0]. Only the transition produces a large value.

A dark-to-bright transition is positive when moving right. Reversing the row gives a negative transition. A derivative is a signed response, not a grey-level image.

Actual filters combine several nearby pixels to obtain a more stable estimate in a two-dimensional image.

Forward difference: $dI/dx \approx I(x + 1, y) - I(x, y)$

Sobel combines difference and smoothing

Gx measures brightness change when moving right; Gy measures change when moving down. Sobel combines a weighted difference with smoothing.

For three identical rows [40, 40, 200], Gx = (200 − 40) × (1 + 2 + 1) = 640; Gy = 0. Derivative responses can exceed 255.

3 × 3 kernel	Top row	Middle row	Bottom row
Gx	−1, 0, +1	−2, 0, +2	−1, 0, +1
Gy	−1, −2, −1	0, 0, 0	+1, +2, +1

Read signed Gx and Gy images



Red is positive; blue is negative; grey is near zero. Both panels use the same scale.

Gx responds to many vertical structures and Gy to many horizontal structures. The labels name the direction of brightness change, not the direction along the edge.

Magnitude and direction

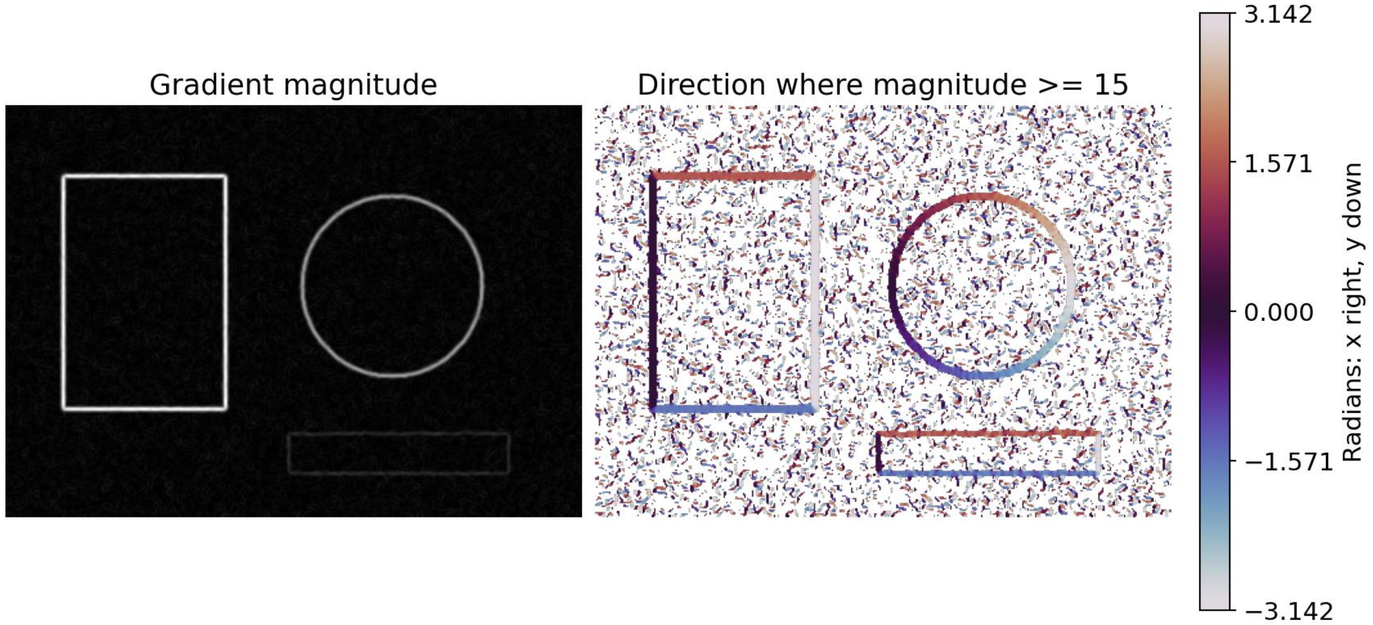
The gradient combines Gx and Gy. Its magnitude describes the strength of local brightness change. Its direction points towards increasing brightness and is perpendicular to the local edge.

Example: Gx = 30 and Gy = 40 give magnitude 50 and direction 53.13 degrees. atan2 uses both signs and handles Gx = 0 safely.

In a nearly flat area, very small responses make direction unstable. A colourful direction there would be misleading.

$$M = \sqrt{(Gx^2 + Gy^2)} \qquad \theta = \operatorname{atan2}(Gy, Gx)$$

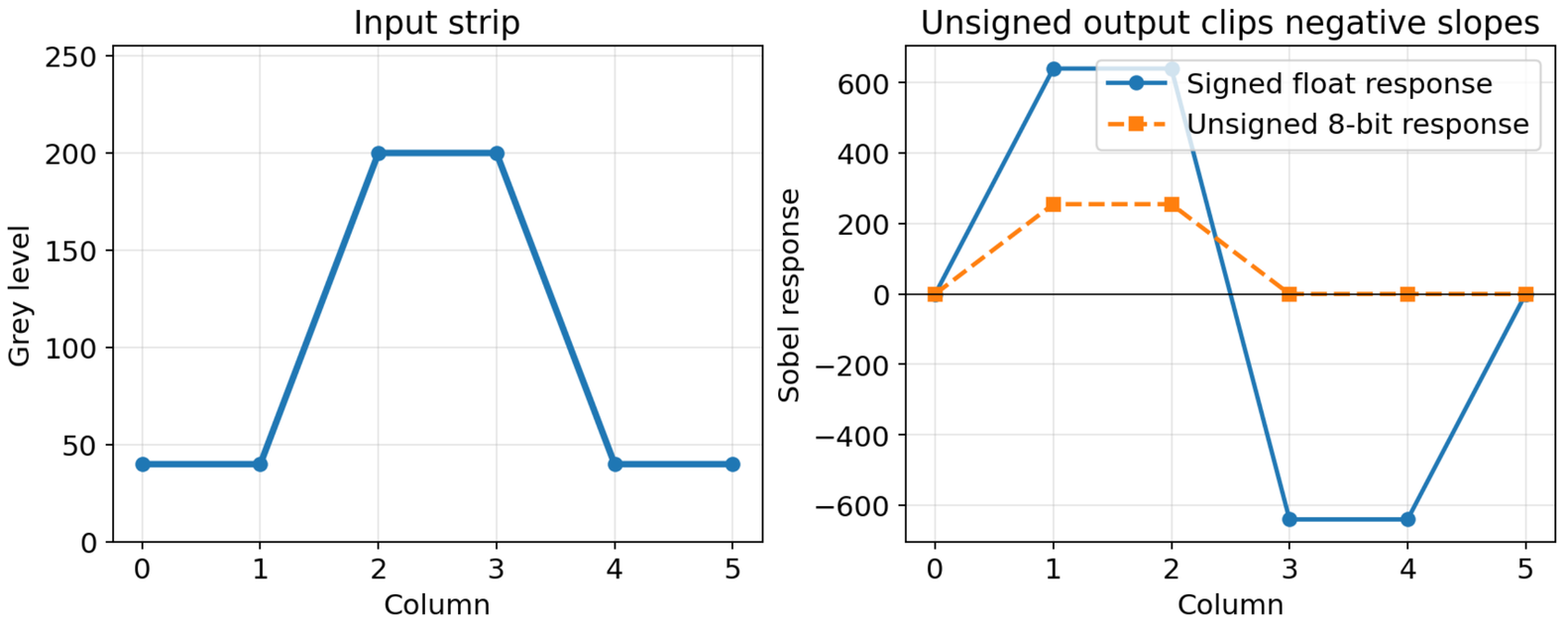
The circle has many gradient directions



The straight sides of a rectangle have mostly constant direction. The circle changes direction around its outline.

The direction map hides responses below 15, so nearly flat regions are blank. Colours describe the direction of local change; they do not label different objects.

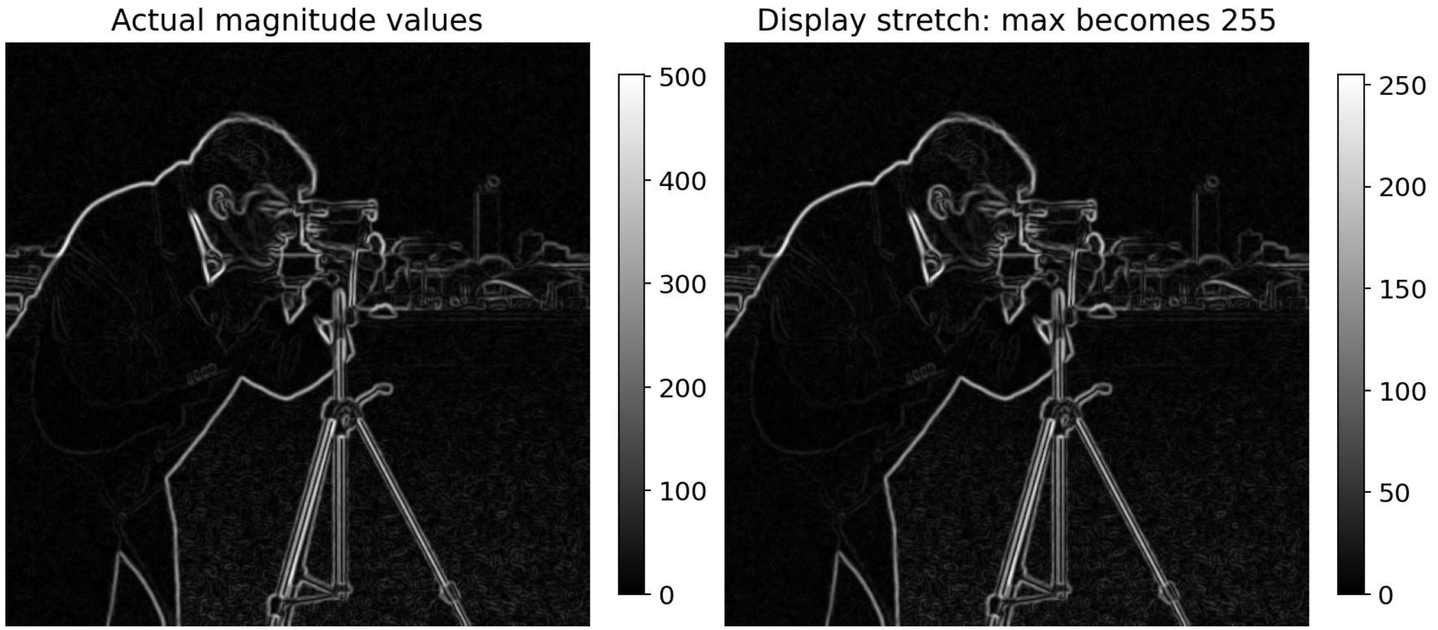
Preserve negative values



Signed Sobel preserves +640 at the rising transition and −640 at the falling transition.

Unsigned 8-bit output clips negative responses to zero and saturates large positives at 255. Store derivatives in floating point first; discard the sign only when it is no longer needed.

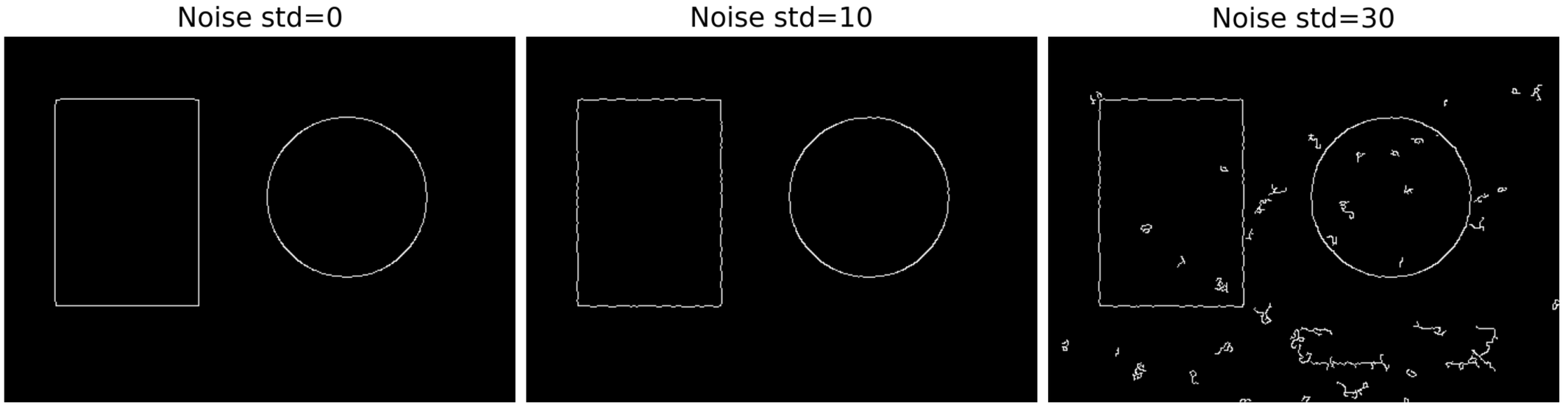
Display normalisation changes the scale



Both panels show the same magnitude array. Its actual maximum is 501.86; the display version maps it to 255.

Normalisation improves visibility but changes the numerical scale. Use raw arrays for calculations. A stretched preview is not a Canny threshold lookup table.

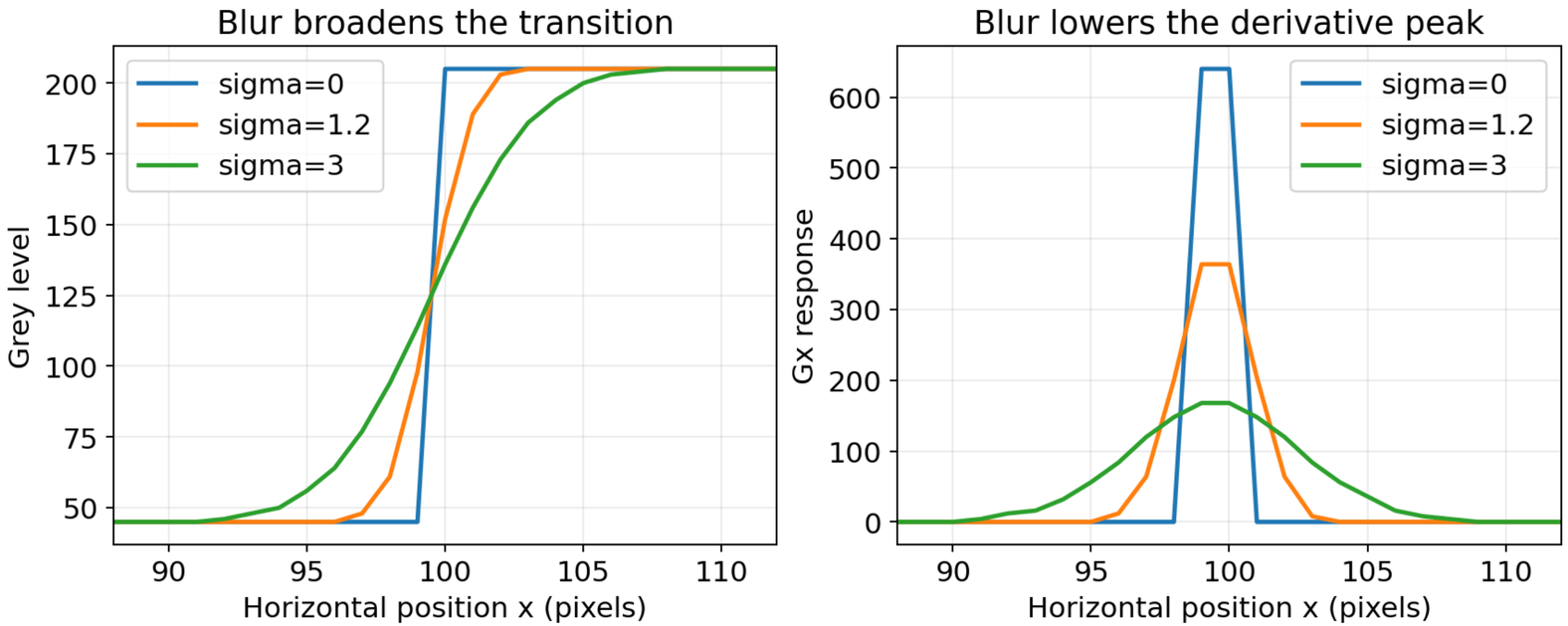
Noise also creates local changes



Random brightness changes create gradients inside uniform areas. More noise can therefore create unwanted edges.

Thresholds and smoothing are identical here. A fixed seed makes the noise repeatable. A more crowded edge map is not automatically better.

Gaussian smoothing sets the scale



Gaussian smoothing averages nearby values with larger weights near the centre. Sigma controls the spread in pixels.

Increasing sigma reduces rapid fluctuations, but also broadens the transition and lowers the gradient peak. Fine detail may disappear. Thresholds and smoothing must therefore be interpreted together.

The Laplacian uses a second derivative

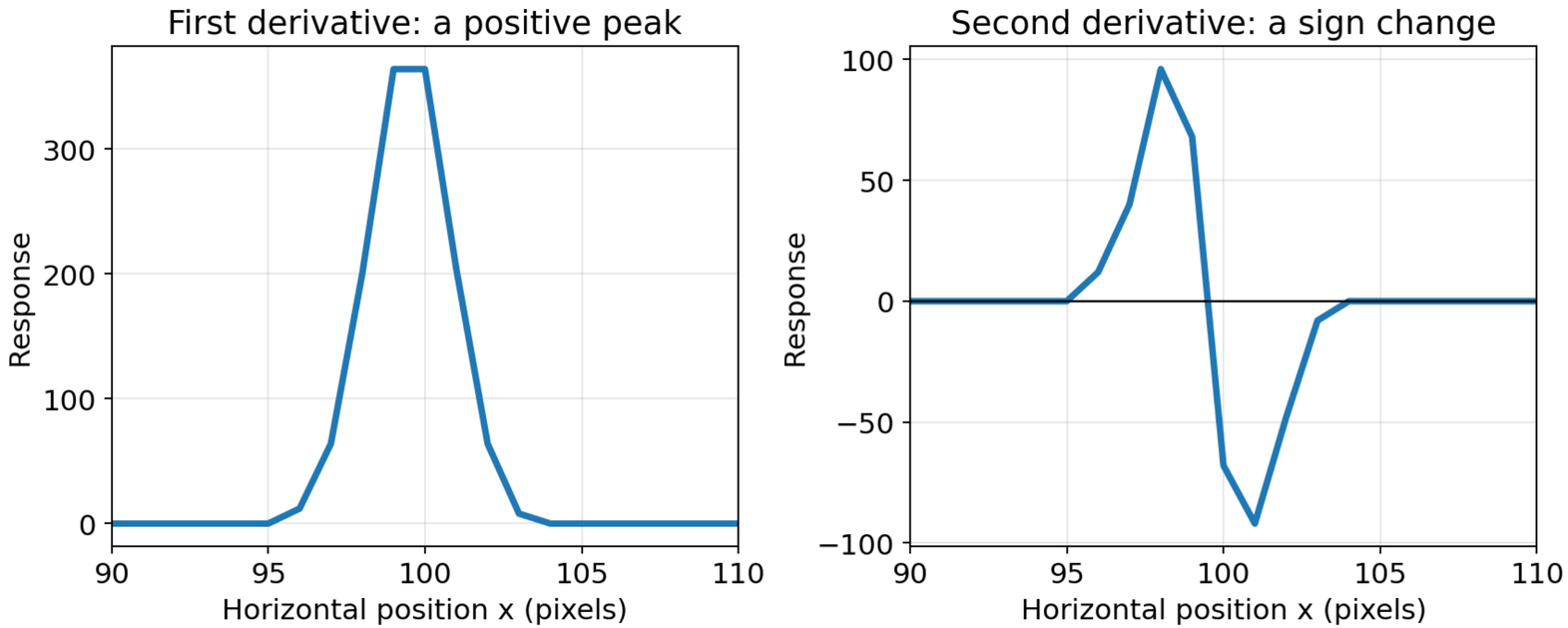
The Laplacian adds the second derivatives in x and y. It describes how the first derivative changes, rather than the direction of brightness increase.

Near a smoothed step, the signed response changes from positive to negative. A zero crossing can indicate an edge location. Flat areas also contain zeros, so a zero value alone is insufficient.

Second derivatives are sensitive to noise. Smoothing and a minimum response change help reject insignificant crossings.

$$\nabla^2 I = \partial^2 I/\partial x^2 + \partial^2 I/\partial y^2$$

A peak and a sign change

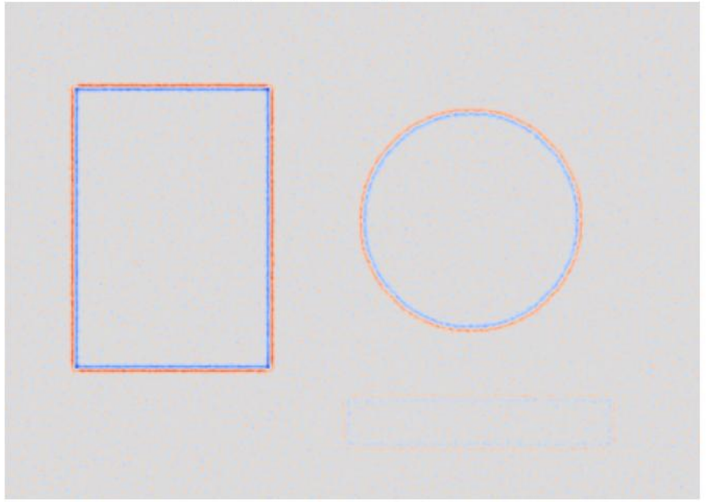


The first derivative peaks near the transition. The second derivative has positive and negative lobes with a sign change between them.

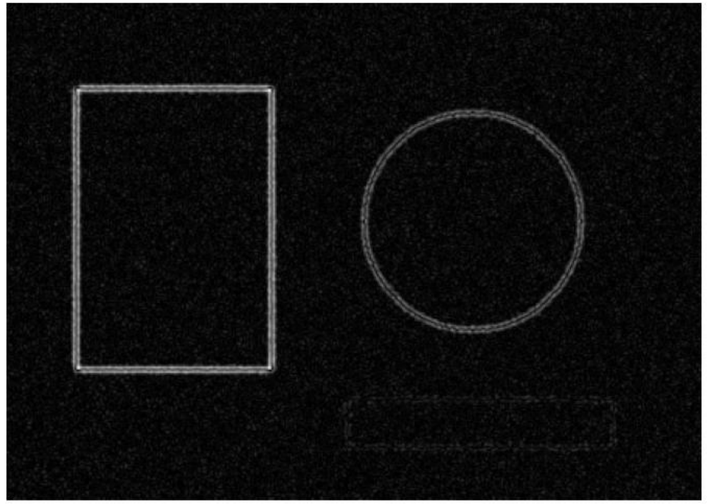
These describe different properties of the same profile. A discrete zero crossing can lie between samples without any response being exactly zero.

Absolute Laplacian is not a zero crossing

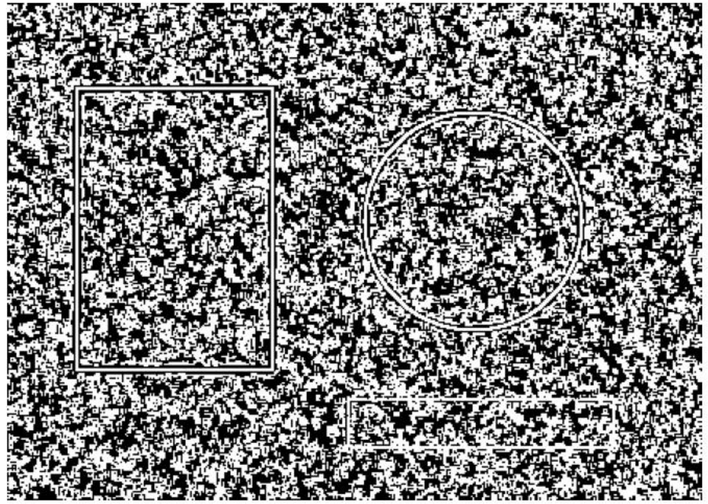
Signed Laplacian



Absolute response: two sides



Sign-change candidates: a band



Absolute values make both Laplacian lobes bright and remove the signs needed to locate a crossing.

The candidate mask tests for both signs in a 3 × 3 neighbourhood, with a response range of at least 20. It produces a band, not a thin contour.

Three methods answer different questions

Sobel and Laplacian produce numerical response images. Extra decisions are needed to turn them into selected edge pixels.

Canny adds thinning, two thresholds and connectivity. Its binary output is useful for outlines, but it still responds to brightness structure rather than object identity.

Method	Direct output	Useful interpretation
Sobel	Signed Gx and Gy	Strength and direction of change
Laplacian	Signed second derivative	Change of slope; possible zero crossings
Canny	Binary edge map	Selected thin, connected edge candidates

The Canny pipeline

1. Smooth the image to reduce noise while keeping relevant structure. 2. Calculate gradient strength and direction.

3. Apply non-maximum suppression: keep local peaks across the edge. 4. Use a low and a high threshold to classify candidates.

5. Retain strong pixels and weak pixels connected to them. Changing thresholds cannot recover detail already removed by heavy blur.

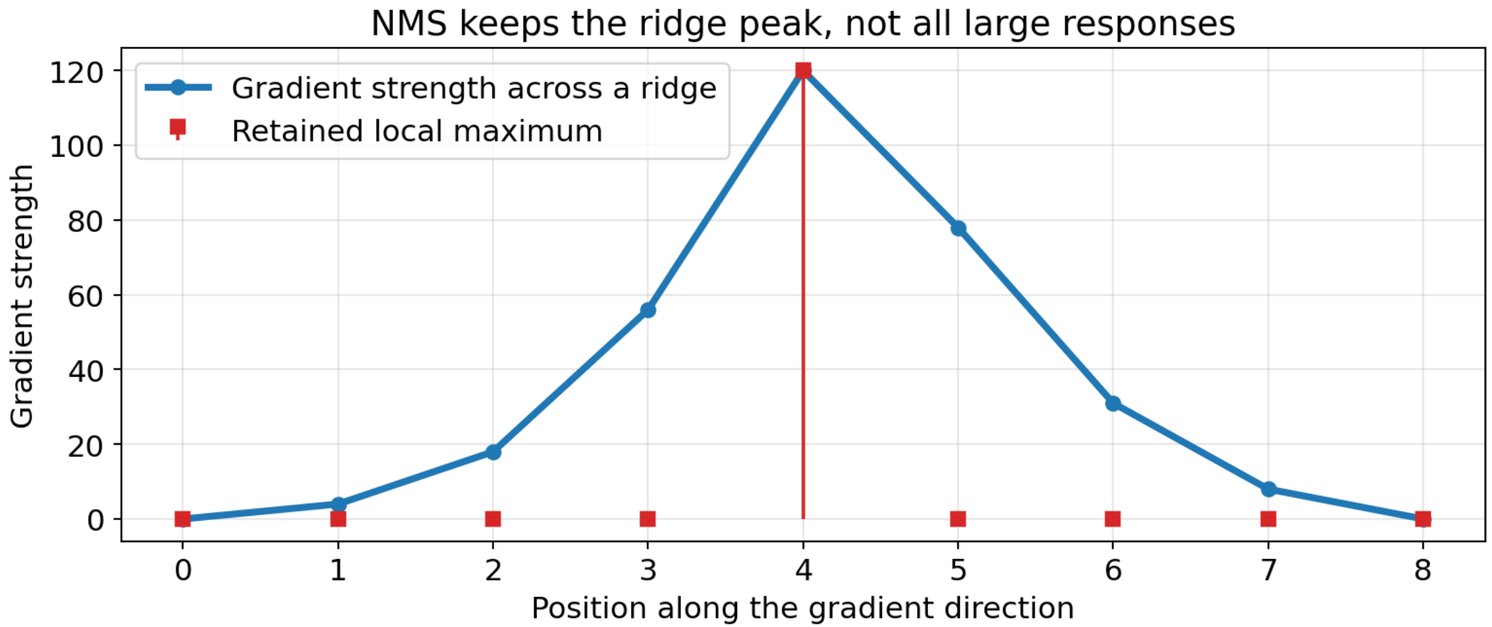
Make the smoothing step explicit



The images show the same noisy photograph after different amounts of Gaussian blur. Noise decreases as small details become softer.

Use GaussianBlur before calling cv2.Canny. The OpenCV Canny function calculates derivatives, thinning and hysteresis; the input should already have the smoothing that the experiment requires.

Non-maximum suppression makes edges thin



Compare responses with their neighbours along the gradient direction, across the edge. Keep the local peak; suppress smaller responses beside it.

Here, 120 survives while 56 and 78 are removed. A large value is insufficient without a local maximum.

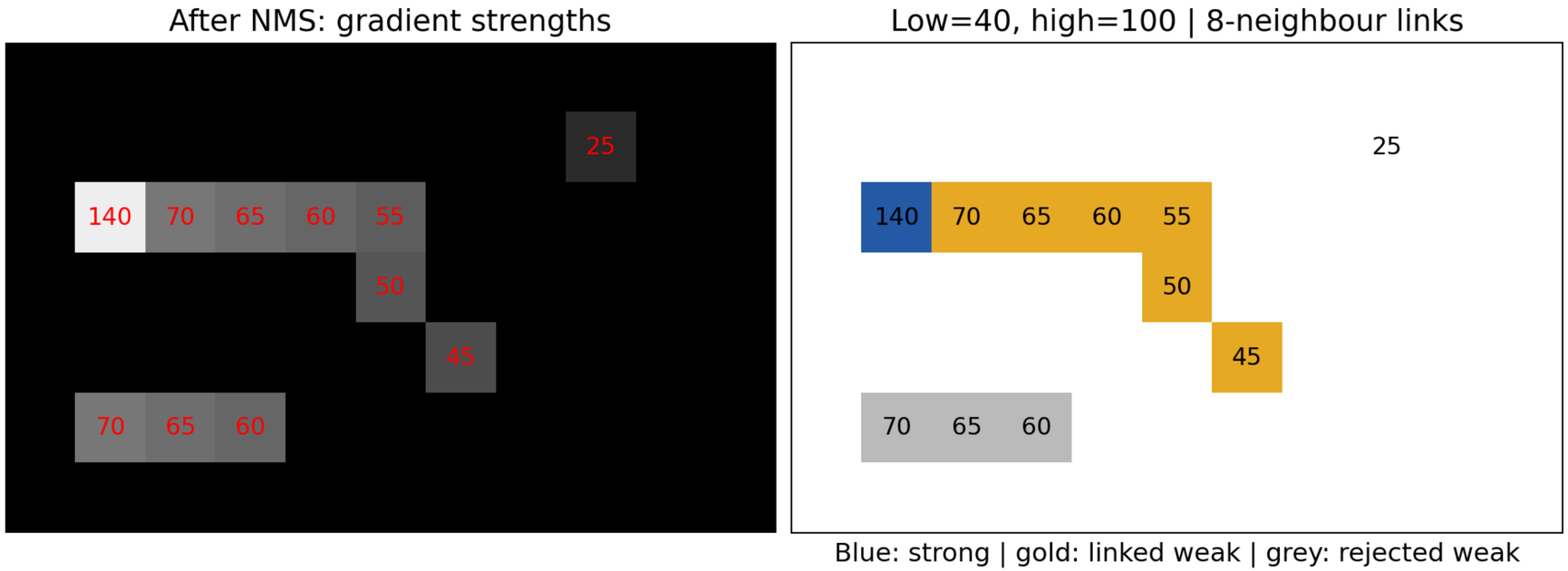
Two thresholds have different roles

After thinning, a response above the high threshold is a strong candidate. A response below the low threshold is rejected. Values between them are weak candidates.

Weak candidates are not accepted independently. Their final result depends on an edge path to a strong candidate. Thresholds act on derivative strength, not directly on input brightness.

Example strength	Low = 40, high = 100	Decision before linking
20	Below low	Reject
70	Between thresholds	Weak: test connectivity
130	Above high	Strong: retain

Hysteresis follows a connected path



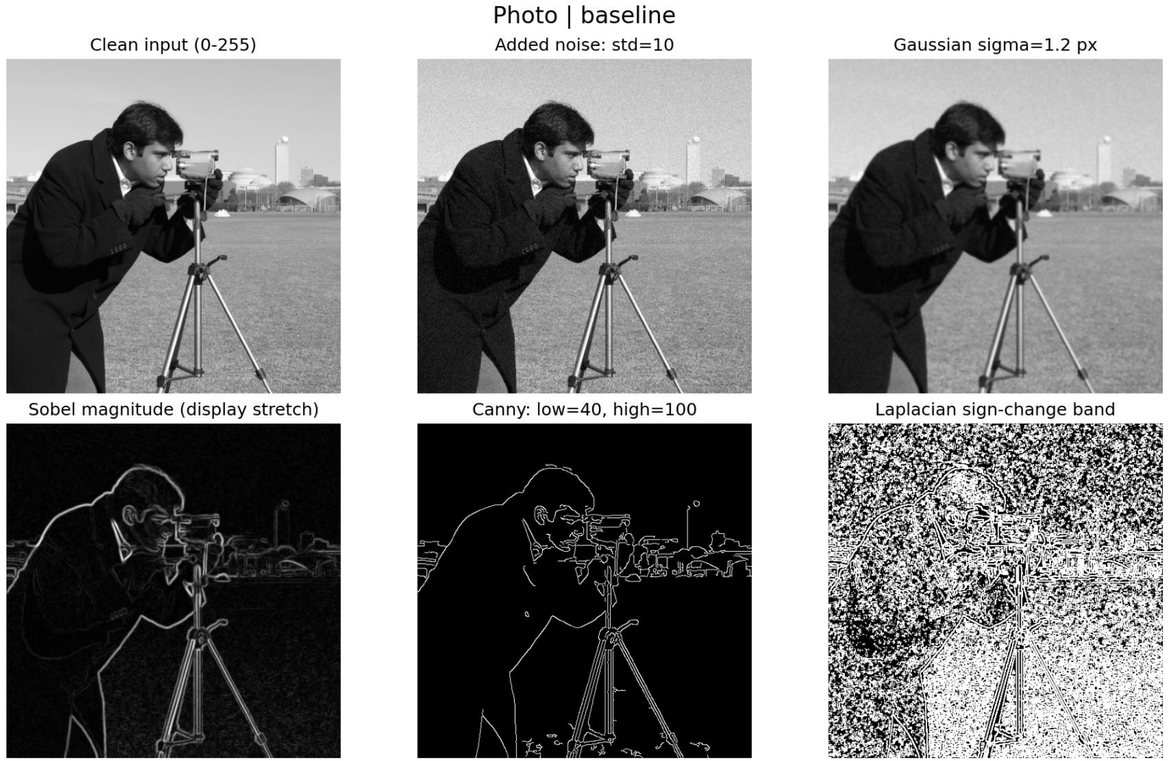
The strong response 140 keeps its connected weaker chain. Diagonal links count with eight neighbours.

The isolated chain 70–65–60 is rejected because it has no strong seed. The response 25 is below the low threshold and cannot bridge a gap.

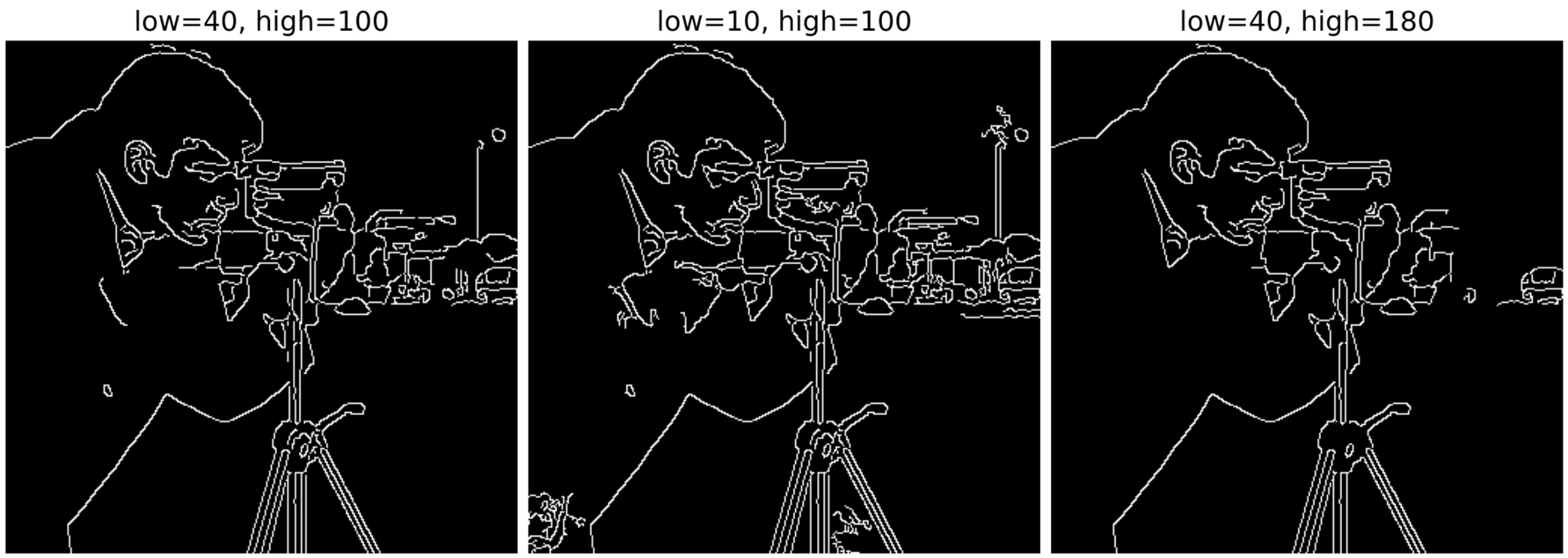
A complete photograph result

Baseline settings: noise std 10, Gaussian sigma 1.2, Canny 40/100 and the L2 gradient norm.

Inspect the person, tripod and background. Sobel shows broad responses; Canny selects thinner structures. The Laplacian sign-change mask is a separate result and can contain wider bands.



Change one threshold at a time



Lowering only low from 40 to 10 admits more linked weak responses: photo count rises from 8,599 to 13,077.

Raising only high from 100 to 180 reduces strong seeds: 6,551 pixels remain. Counts describe selection, not whether selected structures are useful.

A controlled noise experiment

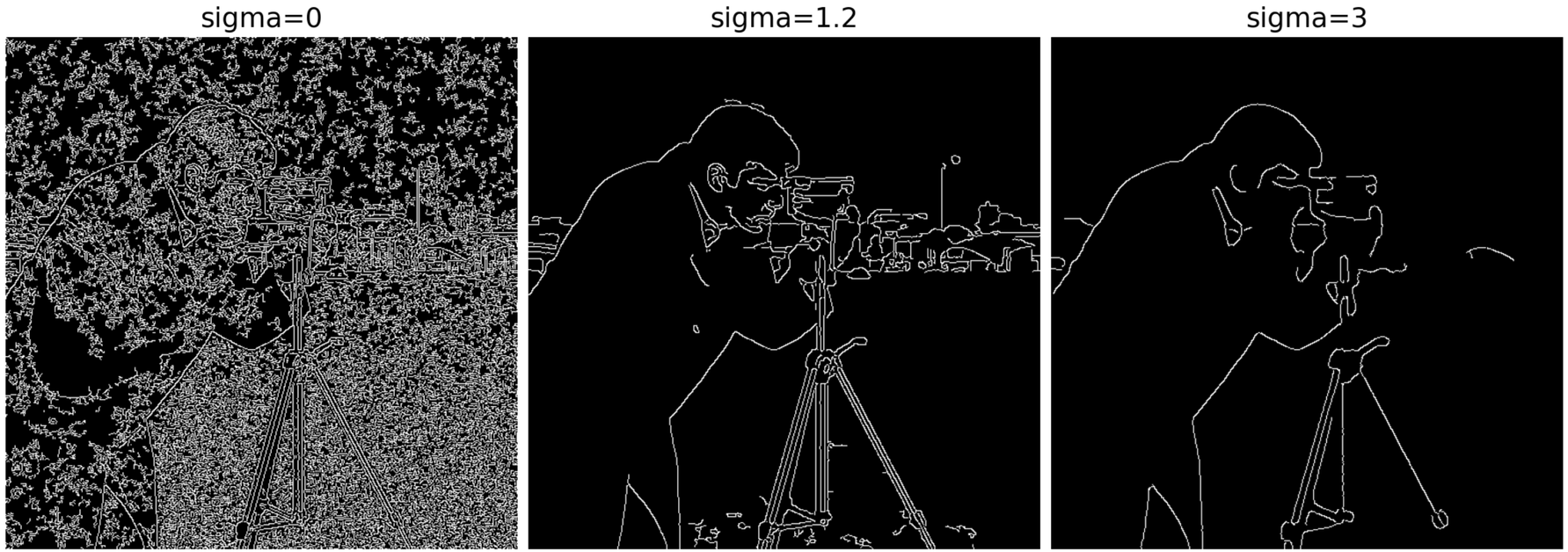
The synthetic input has known boundaries, so extra responses can be checked against a reference. At noise std 10, boundary precision is 1.000 and recall is 0.745.

At noise std 30, precision falls to 0.562 and recall rises to 0.893. Noise can reveal some weak boundaries while adding false structure. One changing number cannot describe this tradeoff.

These rates allow a two-pixel matching distance. They describe this synthetic scene and these settings, not all photographs.

Added noise std	Detected pixels	Boundary precision	Boundary recall
0	1,098	1.000	0.745
10	1,127	1.000	0.745
30	2,525	0.562	0.893

Smoothing removes noise and detail

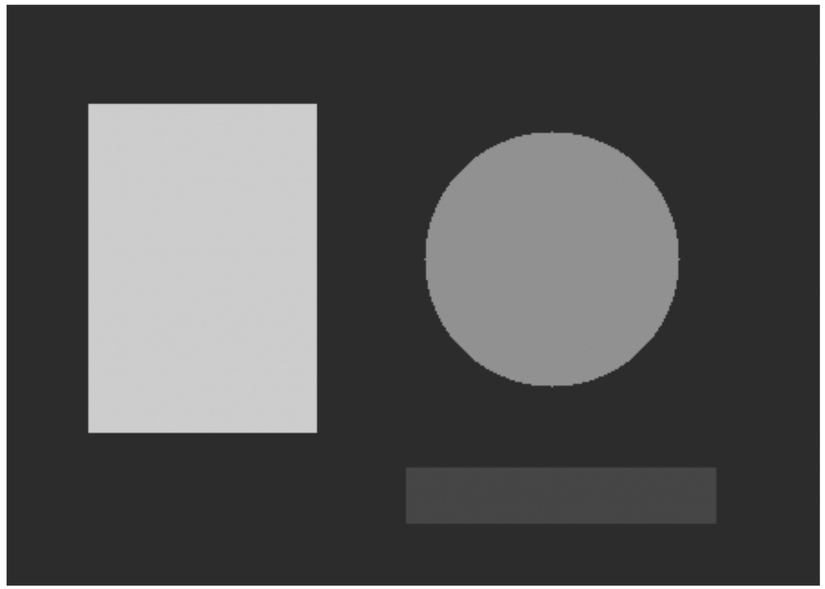


Without smoothing, noise creates many background responses. Moderate blur keeps more useful structure; stronger blur also removes fine detail.

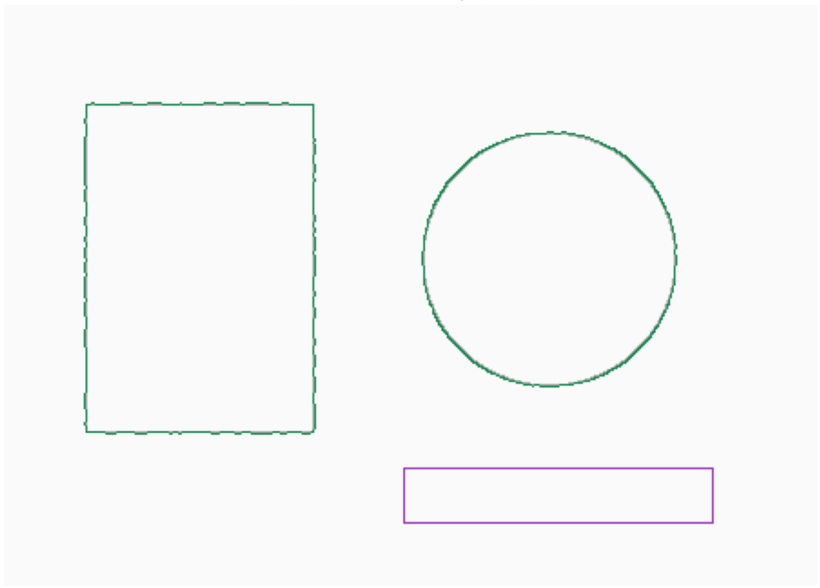
Counts are 67,642, 8,599 and 3,637. No photo boundary reference exists, so counts cannot select the best setting. Inspect the structures your task needs.

Known truth makes errors visible

Synthetic scene: known shape boundaries



Green: matched prediction | purple: missed truth



The reference is the one-pixel inside boundary of the drawn shapes. Matching allows a two-pixel distance.

Green marks matched predictions; purple marks missed reference pixels. The weak lower rectangle is largely absent. A clean-looking map can still miss relevant structure.

Read precision and recall together

Boundary precision asks: what fraction of predicted pixels lie near the reference? Boundary recall asks: what fraction of reference pixels have a nearby prediction?

The two rates use separate many-to-many distance tests. They are not a strict one-to-one boundary benchmark, and they are not whole-region segmentation accuracy.

F1 combines precision and recall, but still depends on the chosen reference and tolerance. Always report that tolerance with the rates.

Precision = matched predictions / all predictions
Recall = found reference pixels / all reference pixels
F1 = 2 × precision × recall / (precision + recall)

More pixels can mean a much worse result

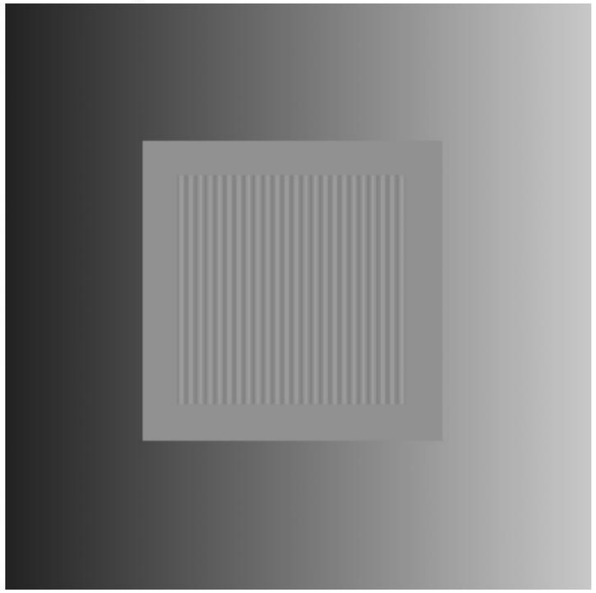
Without smoothing, the synthetic count reaches 23,831, but precision is only 0.082. Many selected pixels do not correspond to drawn boundaries.

Several settings give the same synthetic recall. That is a useful finding: changing a parameter does not guarantee a visible benefit, and one simple scene cannot test every failure mode.

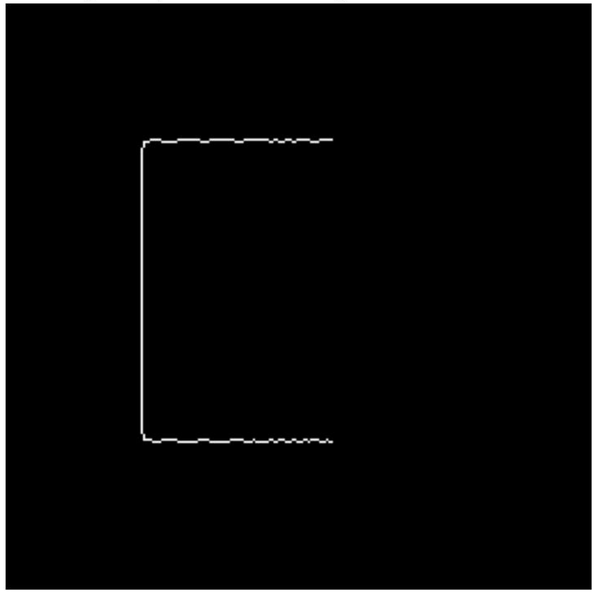
Run	Edge pixels	Precision	Recall
baseline	1,127	1.000	0.745
no smoothing	23,831	0.082	1.000
strong smoothing	1,117	1.000	0.745
low threshold	1,127	1.000	0.745
high threshold	1,127	1.000	0.745
no noise	1,098	1.000	0.745
high noise	2,525	0.562	0.893

Brightness structure is not object meaning

Synthetic object + texture + illumination ramp



Canny responds to brightness structure



The synthetic object contains texture; its background has gradual illumination change. Canny selects brightness structure, including texture.

A physical boundary can be weak when both sides have similar brightness. Strong texture need not be a boundary. Object understanding requires more evidence than edges.

Parameter values need a defined scale

Sigma is in pixels. A different image resolution changes the physical scale per pixel; keep resolution fixed for comparisons.

Canny thresholds act on gradients. Their useful range depends on kernel, norm, contrast and blur. This implementation fixes a 3 × 3 derivative and L2gradient=True.

L2 uses $\sqrt{(Gx^2 + Gy^2)}$; the default L1 norm uses $|Gx| + |Gy|$. Identical thresholds do not give equivalent decisions under both norms.

A practical tuning process

Define the structures that matter: a document outline, a part boundary or small details. Inspect the input at its working resolution.

Choose smoothing that reduces unwanted variation while keeping those structures. Adjust high to keep reliable seeds, then low to connect useful weaker responses.

Change one setting at a time. Compare the same crop and record settings. Use reference boundaries when available; otherwise describe missed structures and unwanted responses.

Review questions with explanations

Why can vertical edges have large Gx? Brightness changes horizontally across them; gradient and edge directions are perpendicular.

Why can weak responses survive? A path above the low threshold can link them to a strong response.

Why is count insufficient? Noise and texture add responses. Inspect relevance, position and continuity.

Why preserve signed derivatives? Negative slopes carry information. Unsigned storage can discard them before magnitude or direction is calculated.